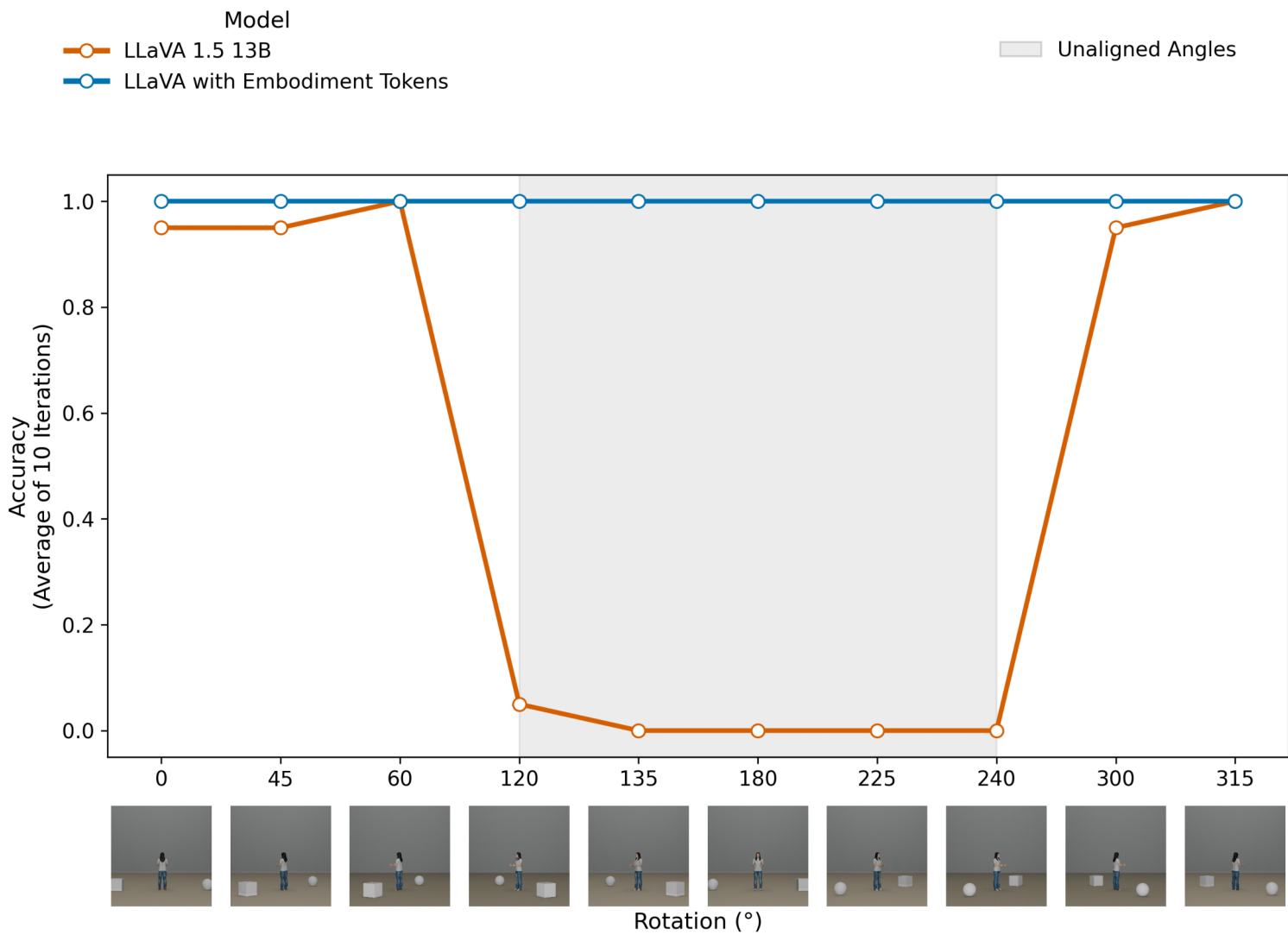
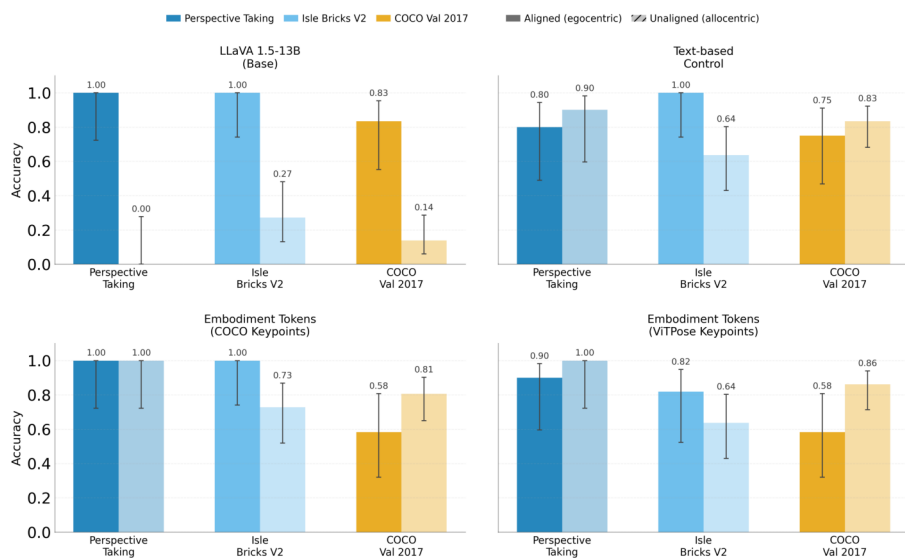


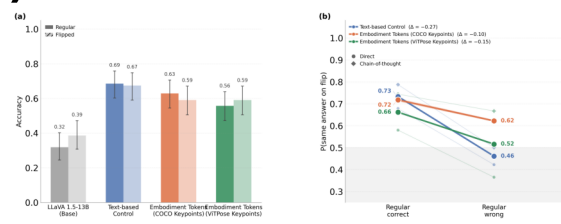
(a)



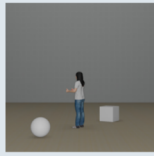
(b)



(c)



(d)

Input (aligned item)	Input (unaligned item)
 Which side of the person is the cube on from the person's perspective?	 Which side of the person is the cube on from the person's perspective?
LLaVA 1.5 13B (Base) From the person's perspective, the cube is on their right side.	LLaVA 1.5 13B (Base) From the person's perspective, the cube is on their right side.
LLaVA with Emb. Text The cube is right of the person from the viewer's frame. The person's pose is 162.0 173.0 [...] 7 0. Since 7 lies in the 315-360 bin, the person is aligned. After performing a mental rotation of 315-360 degrees, the cube is now to the right. Therefore, the cube is to the right from the perspective of the person.	LLaVA with Emb. Text The cube is left of the person from the viewer's frame. The person's pose is 172.0 173.0 [...] 4 1. Since 4 lies in the 180-225 bin, the person is unaligned. After performing a mental rotation of 180-225 degrees, the cube is now to the right. Therefore, the cube is to the right from the perspective of the person.
LLaVA with Emb. Tokens The cube is right of the person from the viewer's frame. The person's pose is <POSE_START> <SHOULDER_L><X_157> <Y_176> [...] <YAW_0> <TORSO_W_1> <ORIENT_END><POSE_END>. Since <YAW_0> lies in the 0-45 bin, the person is aligned. After performing a mental rotation of 0-45 degrees, the cube is now to the right. Therefore, the cube is to the right from the perspective of the person.	LLaVA with Emb. Tokens The cube is right of the person from the viewer's frame. The person's pose is <POSE_START> <SHOULDER_L><X_175><Y_171> [...] <YAW_4> <TORSO_W_1><ORIENT_END><POSE_END>. Since YAW_4 lies in the 180-225 bin, the person is unaligned. After performing a mental rotation of 180-225 degrees, the cube is now to the left. Therefore, the cube is to the left from the perspective of the person.